

Theorem: If $|G| = p^n$, where p is a prime then $|Z(G)| \geq p$.

Proof:

Let $|Z(G)| = r$.

By class equation:

$$|G| = |Z(G)| + \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right| \quad (\text{A})$$

$a \notin Z(G) \implies N(a) \neq G$

$\implies N(a)$ is a proper subgroup of G .

By Lagrange's theorem, $|N(a)| \mid |G|$

$$\implies |N(a)| = p^{\alpha_a}, \quad \alpha_a < n$$

$$\left| \frac{G}{N(a)} \right| = \frac{|G|}{|N(a)|} = \frac{p^n}{p^{\alpha_a}} = p^{n-\alpha_a}, \quad n - \alpha_a > 0$$

$$\implies p \mid \left| \frac{G}{N(a)} \right|$$

$$\implies p \mid \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right|$$

Also, $p \mid |G|$ (as $|G| = p^n$)

$$\implies p \mid \left(|G| - \sum_{a \notin Z(G)} \left| \frac{G}{N(a)} \right| \right)$$

From (A):

$$\implies p \mid |Z(G)|$$

Since $e \in Z(G) \implies |Z(G)| \geq 1$.

$\therefore p \leq |Z(G)| \implies |Z(G)| \geq p$.

$\text{Cl}((12)) = \{(12), (13), (14), (23), (24), (34)\}$

$\text{Cl}((123))$

Theorem: A group of order p^2 , p is prime, is abelian.

Proof:

Let $|G| = p^2$.

$\Rightarrow |Z(G)| \geq p$.

So, $|Z(G)| = p$ or p^2 .

If $|Z(G)| = p$, then

$$\left| \frac{G}{Z(G)} \right| = \frac{p^2}{p} = p$$

$\Rightarrow G/Z(G)$ is cyclic

$\Rightarrow G$ is abelian

$\Rightarrow G = Z(G) \implies |G| = |Z(G)| = p^2$.

$\therefore |Z(G)| \neq p$ (contradiction).

So, $|Z(G)| = p^2 = |G|$

$\Rightarrow G = Z(G)$

$\therefore G$ is abelian.

Note: If $H \leq G$ s.t. $H \leq Z(G)$ then $H \trianglelefteq G$.

Theorem: Let σ be a permutation in S_n and let $m_1, m_2, \dots, m_r$ be the distinct integers which appear in the cycle type of σ (including 1-cycles). Assume that σ has k_i cycles of length m_i for each $i = 1, 2, \dots, r$ so that $\sum_{i=1}^r k_i m_i = n$.

Then number of conjugates of σ is

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \cdots (k_r! m_r^{k_r})}$$

Example: In S_4

No. of cycles of 2 length in S_4 :

$$\frac{4!}{(1! 2^1)(2! 1^2)} = \frac{24}{2 \times 2} = 6$$

No. of cycles of length 3:

$$\frac{4!}{(1! 3^1)(1! 1^1)} = \frac{24}{3} = 8$$

In S_5 :

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$
$$(3! 1^3)(1! 2^1) = 6 \times 2 = 12 \implies \frac{120}{12} = 10$$

Ring: A system $(R, +, \cdot)$ consisting of a non-empty set R and two internal binary operations ‘+’ and ‘ $\cdot$ ’ in R is called a **ring** if:

1. $(R, +)$ is an abelian group
2. $(R, \cdot)$ is a semi-group
3. Distributive laws hold:

$$a \cdot (b + c) = a \cdot b + a \cdot c$$
$$(a + b) \cdot c = a \cdot c + b \cdot c \quad \forall a, b, c \in R$$

If $(R, \cdot)$ has identity $\implies$ **Ring with unity**.

If $(R, \cdot)$ is abelian $\implies$ **Commutative ring**.

Note:

1. ‘+’ is usually called addition and ‘ $\cdot$ ’ is usually called multiplication.
2. Identity of $(R, +)$ is called the **zero element** of ring and is denoted as 0.
3. The identity w.r.t. ‘ $\cdot$ ’ is known as **unity** (identity element) of the ring and is denoted as 1.

4. If ‘ $\cdot$ ’ is commutative then R is called as **commutative ring**.
5. If the identity exists in R then R is called as **ring with identity** or **ring with unity**.

Example:

1. $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ all are commutative ring with unity.
2. $(\mathbb{Z}_n, +, \cdot)$ is a commutative ring with unity.
3. $\mathbb{Z}[\sqrt{p}] = \{a + b\sqrt{p} \mid a, b \in \mathbb{Z}, p \text{ is prime}\}$. Then it is a commutative ring with unity.
4. $\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$ is also a commutative ring with unity (**Ring of Gaussian Integers**).
5. $M_n(\mathbb{R}) =$ set of all $n \times n$ matrices with real entries.
 $(M_n(\mathbb{R}), +)$ is group.
 $(M_n(\mathbb{R}), \cdot)$ is a semi-group.
 $A \cdot (B + C) = AB + AC$
 $(A + B) \cdot C = AC + BC$
 $\therefore (M_n(\mathbb{R}), +, \cdot)$ is a non-commutative ring with unity.